

Darren Buttigieg

Python Developer | Scientific Computing | Backend Systems

Remote · EU timezone · Open to contract & permanent

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About

Engineer with 8+ years of Python experience. My background in astrophysics and mathematics shaped how I approach complex problems, be they research papers, mathematical models, or ambiguous product ideas. Modern AI tooling lets me move fast across stacks, backed by engineering judgment built over nearly a decade to know what correct, maintainable code actually looks like.

My core strength is reasoning from first principles, learning a new domain quickly, and shipping something real — from petabyte-scale data ingestion at [Onna](#), to a multi-year economic index platform built independently, to recent product work like [ClimateMapper](#) and my own [portfolio site](#), both built solo using AI-assisted workflows.

Core Skills

Python 3
Backend Systems / REST APIs
Data Engineering / ETL
Scientific Computing
Docker
GitHub

Languages

Maltese (native)
English (native)
Spanish (very good)
Italian (very good)
French (basic)

Education

MSc Astronomy, Cosmology

Leiden University, Netherlands
2014–2016

Thesis: Exploring the Phenomenology of Dark Energy / Modified Gravity on Cosmological Scales

Python · Matplotlib · NumPy · AstroPy · Pandas · Wolfram Mathematica · LaTeX

BSc (Hons) Mathematics & Physics

Summa Cum Laude
University of Malta
2010–2014

Java · C++ · Fortran · Wolfram Mathematica · MATLAB

Experience

Freelance Software Developer

Nov 2017 – May 2020, Mar 2022 – Present

- Independently evolved from prototype a platform constructing economic indices from large-scale global news datasets, implementing Baker-Bloom-Davis-style academic methodology across tens of thousands of articles spanning multiple decades and news sources
- Built [ClimateMapper](#), a geospatial visualisation tool for exploring climate variables derived from ERA5 reanalysis data, with an interactive map interface for regional and temporal comparison
- Built CouchSearch, a geospatial discovery tool mapping hosts by region with a layered visual encoding system representing availability, references, and activity recency
- Implemented scientific models from academic literature across multiple domains: computational neuroscience (Hodgkin-Huxley neuron dynamics), epidemiology (SIR/SIRV compartmental ODE models), structural and continuum mechanics (FEniCS / FEM — arterial blood flow, PDE-based physical simulations), and predator-prey population dynamics
- End-to-end delivery for 50+ clients: architecture, implementation, client alignment, and deployment
- Accepted into [Toptal network](#) of top 3% talent (November 2022)
- Python 3 · Claude Code · GitHub Copilot · GitHub Actions · Docker · Git · Selenium · Matplotlib · NumPy · Pandas · Wolfram Mathematica · LaTeX

Software Engineer, Connectors Team — [Onna](#)

Jun 2020 – Feb 2022

Barcelona & Remote — 15–20 person engineering team

- Contributed to 20+ third-party API connectors for large-scale backend systems (billion-file / multi-petabyte scale), used by companies including Electronic Arts, Dropbox, and Slack
- Designed data synchronisation logic reducing API calls by 40% and rate limiting by 90%, improving sync time by 1.5x+
- Implemented monitoring and observability (Grafana, Prometheus), identifying all instances and causes of rate limiting and leading to 30+ bugs being caught before production
- Python 3 · SQLAlchemy · Asyncio · Redis · RabbitMQ · Docker · Git · Kubernetes · Jenkins · Grafana · Prometheus

Software Developer — [Sherpa](#)

Oct 2016 – Oct 2017

Malta

- Co-authored a research paper proposing a novel mathematical approach to quantifying and scoring risk across 5+ categories
- Implemented mathematical models in production code, processing 100+ data points per assessment with careful attention to numerical stability and model validity
- Contributed to backend systems exposing scoring and risk models as a client-facing platform
- Python 3 · Flask · Swagger · SQLAlchemy · Git